

AN UNSUPERVISED ALIGNMENT FEATURE FUSION SYSTEM FOR SPOKEN LANGUAGE-BASED DEMENTIA DETECTION

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ABSTRACT

Alzheimer's Disease (AD) is a neurodegenerative disorder characterized by irreversible cognitive decline. As no cure, early detection is crucial for timely treatment. In the early stages, AD often manifests through impairments in spoken language, affecting both acoustic and linguistic abilities. Previous research has shown promising results in developing spoken language-based AD detection systems, but lacking a consistent and effective strategy for fusing the acoustic and linguistic features. In this study, we introduce a novel feature fusion framework which considering the correlation between these two modalities. To achieve this, prior to fusion, we design a unsupervised alignment mechanism to learn the pairwise affinity between the two modalities. Our framework is evaluated on the English (ADReSS and DementiaBank) and Chinese (MCGD) datasets, achieving promising results on all of them. Additionally, we visualize the learned alignment matrix, highlighting the correlation between the speech and language in AD detection. These findings offer valuable insights for future research and potential clinical applications.

Index Terms— Soft Alignment, Alzheimer's Disease Detection, Multimodal Fusion, Pretrained Language Model

1. INTRODUCTION

Alzheimer's disease (AD), a neurodegenerative disorder, causes long-term and irreversible cognitive decline. Given the lack of effective treatments for AD, early diagnosis is critical for enabling timely intervention and slowing disease progression. Though memory deterioration is the main symptom, acoustic and linguistic abilities decline is also sensitive to AD at its early stages. For the acoustic part, the movement of the articulators is affected by the development of AD [1]. For the linguistic part, the memory deficits, attention deficits, and visual perceptual problems resulting from

AD can all be affected [2]. With these research, constructing a speech-and-language based AD detection system is promising for automatic AD detection at the early stages.

Inspired by the clinical diagnosis, where both the acoustic and linguistic information are considered by clinicians, the acoustic and linguistic feature fusion system is expected to be more efficient than the unimodal AD detection system. However, while constructing an AD detection system, not all the multimodal AD detection systems can guarantee a better performance compared to the unimodal models [3]. It is inferred that the correlation between the two modalities is ignored while doing feature fusion. In the general multimodal feature fusion research fields, feature alignment has been demonstrated to improve the following-up feature fusion efficiency [4], which encourage us to utilize feature alignment before fusion while designing an AD detection system.

In this work, we propose a novel feature fusion framework of aligning the text and speech embeddings extracted with BERT and Whisper to capture the correlation information between different modals. Our proposed system achieved an F-score of 91.30% on the ADReSS dataset, 91.43% on the DementiaBank dataset, and 80.65% on the MCGD dataset. The visualized speech and text aligned matrix learned with the aligning mechanism reveals the speech and language pattern difference between people living with and without AD.

2. RELATED WORK

In previous research, both the acoustic and linguistic based AD automatic detection has been explored. With the development of pre-trained language models, Whisper and BERT have been shown to effectively learn both acoustic and linguistic information for dementia detection. For instance, AD-BERT system was proposed to utilize the pre-trained BERT model for linguistic-based AD detection [5]. Li et al. introduced a method where the encoder part of Whisper is frozen, and its decoder is fine-tuned on the ADReSSo dataset to extract acoustic information [6]. To effectively utilize

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both acoustic and linguistic information, feature embeddings from audio recordings and transcripts are typically fused through concatenation or cross-attention when constructing a multimodal system [7, 8]. However, in some cases, the performance of the fused system does not surpass that of a single-modality system [3]. It is inferred that repetition and pause patterns, which are significant for AD detection [9, 10], remain misaligned in the shared feature space between audio recordings and the corresponding transcripts.

In data mining and cross-modal retrieval, learning semantic alignment between speech and text has demonstrated effective for enhancing joint representation learning. For downstream tasks such as speech-to-text translation and text-to-speech (TTS) synthesis, alignment mechanisms aim to minimize the distribution gap between speech and text embeddings in shared latent spaces. In [11], the HMM-based Viterbi-forward-backward framework learns unsupervised alignment between mel-spectrograms and phoneme sequences by optimizing the maximum likelihood path, eliminating the need for forced-alignment labels for TTS. Inspired by [11], in this work, an unsupervised alignment strategy was proposed for speech-text feature fusion in AD detection, to leverage the robustness to pathological speech variations and capability to operate without manually annotated boundaries.

3. METHODOLOGY

In this section, we provide a detailed description of our proposed system. The system architecture is illustrated in Figure 1. As shown, the acoustic and linguistic feature embeddings are learned using Whisper and BERT, respectively, and the two embeddings are subsequently used to learn the alignment matrices for feature fusion.

3.1. Feature Embedding Extraction

Let S be a piece of speech composed of an order of phonemes $S_1, S_2, \dots, S_L$ along the temporal domain, where L represents the lengths of S . Let T be a piece of text composed of an order of words $T_1, T_2, \dots, T_N$ along the temporal domain, where N represents the lengths of T . The audio recordings are transformed into mel-frequencies before fed into Whisper, followed by a dropout layer. The outputs of the dropout layer are represented as S_{emb} for audio recordings, and T_{emb} for manual transcripts. To learn the high-level representation and reduce dimension of the audio recordings and manual transcripts, three one-dimensional convolutional (Conv1D) layers and two Conv1D layers are utilized for speech and text embeddings (as in [11]). Then S_{emb} and T_{emb} are transformed into S_{high} and T_{high} , respectively.

3.2. Unsupervised Modality Alignment

To obtain the alignment relation between S_{high} and T_{high} , D is defined as the matrix corresponding to the L2 distance

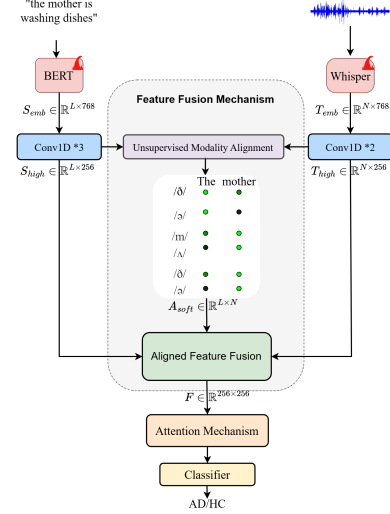


Fig. 1. The structure of the proposed feature fusion system utilizing with the unsupervised feature alignment mechanism. “HC” denotes Health Control.

between S_{high} and T_{high} , while $A_{soft} \in \mathbb{R}^{L \times N}$ is defined as the alignment matrix between speech embedding S_{high} and text embedding T_{high} , as shown in Figure 1. The equation for calculating D and $A_{soft} \in \mathbb{R}^{L \times N}$ are shown as follows:

$$D_{i,j} = \text{distance}_{L2}(T_{high,i}, S_{high,j}) \quad (1)$$

$$A_{soft} = \text{softmax}(-D, \text{dim} = 0). \quad (2)$$

In $D_{i,j}$, each column of D representing pieces of the speech embedding contains a list of distances to the text embedding, which represent the likelihood that the raw records corresponding to these two embeddings are spoken out approximately. Specifically, the smaller the distances between the two embeddings, the higher the likelihood that the raw records occur at roughly the same time and vice versa. To learn the relevance probability, softmax is utilized. In A_{soft} , every row involves the relative alignment relationship from a single speech embedding to every text embedding, and the orders of the rows and columns in the matrix represent the temporal order.

3.3. Aligned Feature Fusion

To fuse the alignment information with the acoustic and linguistic embeddings, matrix multiplication is applied to fuse the embeddings of speech S , alignment A_{soft} , text T in order to compute the fused feature F as follows:

$$F = S^T \times A_{soft} \times T. \quad (3)$$

The fused feature F incorporates alignment information, along with the embeddings derived from audio recordings and manual transcripts. A self-attention mechanism is then

employed to capture the relationships between the alignment information from both preceding and subsequent segments. Following feature fusion, our proposed model is followed by a linear layer with a *logistic* function, which serves as the classifier for classifying HC from AD.

4. EXPERIMENTAL SETUP AND DATASETS

In this section, we first introduce the corpora and implementation details used in our experiments, followed by a description of the data pre-processing methods.

4.1. Dataset Information

The DementiaBank corpus [12] consists of 551 records from 293 speakers. However, some participants experienced changes in their diagnoses as they aged. After removing these samples, the remaining dataset includes 222 samples from 89 healthy controls and 255 samples from 168 AD. The ADReSS dataset [13] is a subset of the DementiaBank corpus, featuring balanced distributions of age and gender. It includes both the original chunked audio recordings and the enhanced full audio recordings. The dataset is divided into a training set and a test set, with the training set consisting of 54 records and the test set containing 24 records from each group. The MCGD dataset [14] was collected from 92 participants (40 male and 52 female) recruited from communities in Shanghai, China. All participants completed a picture description task based on the Cookie Theft picture description and MoCA-B test under the guidance of trained operators. The dataset is divided into 40 AD samples, 40 mild cognitive impairment (MCI) samples and 22 HC samples according to the MoCA-B score. In this experiment, only the samples from HC and AD are used for evaluating our proposed method.

4.2. Data Pre-processing

As proposed by [15], the model trained with the DementiaBank corpus was proposed to inadvertently learn spurious correlations in the data to achieve high performance metrics, a phenomenon known as the "Clever Hans Effect". To mitigate this effect, the audio recordings are enhanced using the method outlined by [16]. The results from both the original and enhanced audio recordings are presented in Section 5. The ADReSS dataset is originally provided in the CHAT format for manual transcripts, which we converted into plain text format. During this process, most special characters and unnecessary punctuation were removed.

4.3. Experimental Setting Details

In this paper, BERT was initialized using *bert-base-cased*, and Whisper was initialized with *whisper-small.en*. Two 768-dimensional embedding vectors were extracted from the last layer of both models. For the speech embedding, three

Conv1D layers with kernel sizes of 3, 3, and 1 were designed for feature dimension reduction. For the text embedding, two Conv1D layers with kernel sizes of 3 and 1 were designed for feature dimension reduction. Our model was trained using the AdamW optimizer with a learning rate of $\eta = 1 \times 10^{-5}$ for 5 epochs. The batch size was set to 1, and the dropout rate was set to 0.4. To evaluate our proposed system, 5-fold cross-validation is utilized on the three dataset.

In this paper, cross validation (CV) is utilized on the datasets for evaluating our designed system. For the DementiaBank dataset, the audio recordings were divided into 5 folds, where 3 folds were used for training, 1 fold for validation, and 1 fold for testing. For the ADReSS dataset, which provides a separate test set, the training set was divided into 5 folds, with 4 folds used for training and 1 fold for validation. The final result is reported with majority voting on the 5 folds. For the MCGD dataset, considering the limited available samples, we employed leave-one-out CV for model evaluation.

5. RESULTS

In this section, we first evaluate the effectiveness of our system by comparing it with the baseline systems we designed and previous research. Then, we analyze the interpretability of our model by visualizing the learned alignment matrix.

5.1. Experimental Results

The results of our designed system on the three datasets, together with the efficiency of the alignment method and attention mechanism is evaluated in this section. The result are listed in Table 1. As shown, both the soft alignment and attention mechanisms play a crucial role in the classification task across different datasets. Specifically, on the ADReSS dataset, the model achieves an accuracy of 76.04% with only the attention mechanism, compared to 89.58% with the soft alignment mechanism alone. When both the alignment and attention mechanisms are applied together, the accuracy increases to 91.67%. A similar trend is observed on the other datasets, further highlighting the effectiveness of soft alignment followed by attention mechanism in feature fusion.

Compared to previous research on the DementiaBank corpus [3, 17–19], our system demonstrates superior performance. Specifically, in [17], an accuracy of 86.65% was achieved by concatenating features from two modalities, which was lower than the linguistic-only system's accuracy of 90.36%. In [3], multiple unimodal and multimodal systems were evaluated. The proposed two-step feature fusion system achieved an accuracy of 81.51%. In [16, 20], an accuracy of 88.96% and 90.24% was achieved on the DementiaBank-E dataset. In contrast, the multimodal feature fusion system with a soft alignment mechanism proposed in this paper outperforms previous systems on the DementiaBank dataset.

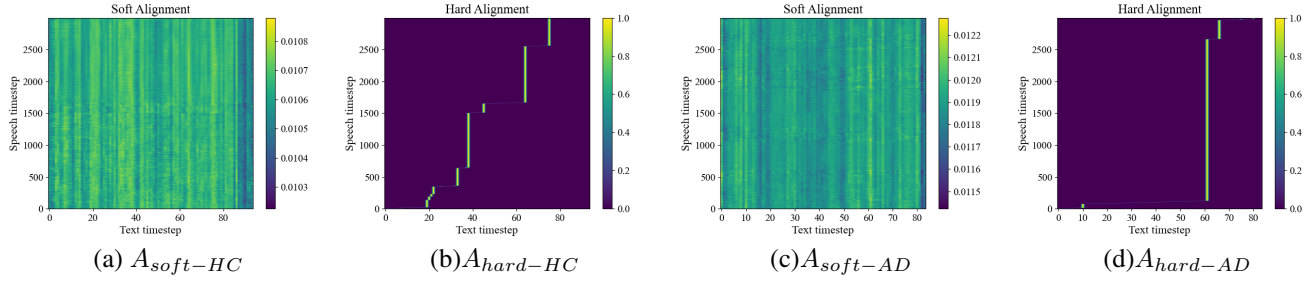


Fig. 2. Alignment Visualization: Fig. (a) and Fig. (b) represent the alignment result of one of the healthy individual; Fig. (c) and Fig. (d) represent the alignment result of one of AD patient.

Table 1. The results (%) of our feature fusion system.

Dataset	Alignment	Attention	ACC	F1
DemBank	✓	✓	83.54	84.54
	✓	✓	85.22	85.54
			87.51	90.85
DemBank-E	✓	✓	86.16	87.43
	✓	✓	85.96	86.02
			90.53	91.43
ADReSS	✓	✓	76.04	76.28
	✓	✓	89.58	88.89
			91.67	91.30
MCGD	✓	✓	67.31	73.85
	✓	✓	69.23	77.78
			76.92	80.65

For the ADReSS dataset, Zhang et al. [18] achieved an accuracy of 89.58% by fusing audio, text, and expert knowledge. In the study proposed by [19], audio recordings were converted into log-Mel spectrograms, and acoustic embeddings were learned using a vision transformer, which were then combined with linguistic features extracted from transcripts using BERT. The best accuracy achieved was 88.33%. In comparison, our multimodal model achieves an accuracy of 91.67% and an F1-score of 91.30%, respectively.

For the MCGD dataset, an F-score of 83.3% was achieved by utilizing the acoustic and linguistic hand-crafted features classified with SVM in [14], compared to 80.65% F-score achieved by our designed system. It is inferred that limited available data for training the proposed end-to-end system.

5.2. Alignment Matrix Visualization

To facilitate a clearer understanding of our proposed unsupervised alignment module, we visualize a representative learned soft-alignment matrix in this section, as depicted in Figure 2. For comparative clarity, we also compute the correspond-

ing hard-alignment matrix. The hard alignment is obtained via monotonic alignment search, which identifies the optimal monotonic alignment path using dynamic programming in the log-probability domain. In our experiments, both soft and hard alignment strategies were evaluated for feature fusion. However, only the soft alignment matrix consistently yielded superior performance. It is inferred that soft alignment can capture global contextual relevance between speech and transcript, which proves more effective for feature fusion.

As shown, Figures 2 (a) and 2 (c) display the soft alignment distances, while Figures 2 (b) and 2 (d) represent the hard alignment results from Figures 2 (a) and 2 (c), respectively. The horizontal axis corresponds to the order of transcript tokens, while the vertical axis reflects the order of speech frames. As seen, the alignment information differs between people living without and with AD. A comparison of Figures 2 (a) and 2 (b) with Figures 2 (c) and 2 (d) reveals that people living with AD tend to linger on individual words or a few consecutive words, while HCs maintain a smoother flow. This observation is consistent with the understanding that individuals with cognitive impairments often struggle with verbal expression, resulting in more frequent pauses at points where HCs would speak fluently.

6. CONCLUSIONS

In this work, we propose a feature fusion model for automatic AD detection, with a particular focus on exploring the alignment between audio recordings and transcripts. Our findings reveal clear differences in speech patterns between HC and individuals with AD. The results demonstrate that incorporating alignment information across these two modalities significantly enhances the performance of the feature fusion system. For future research, we plan to extend the evaluation of our method to other speech-related disorders, such as depression and Parkinson's disease, and explore integration with other advanced architectures, such as large language models.

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